

Identity Confusion in WebView-based Mobile App-in-app Ecosystems

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Identity Confusion in WebView-based Mobile App-in-app Ecosystems

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Distinguished Paper Award Winner

Mobile applications (apps) often delegate their own functions to other parties, which makes them become a super ecosystem hosting these parties. Therefore, such mobile apps are being called super-apps, and the delegated parties are subsequently called sub-apps, behaving like "app-in-app". Sub-apps not only load (third-party) resources like a normal app, but also have access to the privileged APIs provided by the super-app. This leads to an important research question—determining who can access these privileged APIs.

Real-world super-apps, according to our study, adopt three types of identities—namely web domains, sub-app IDs, and capabilities—to determine privileged API access. However, existing identity checks of these three types are often not well designed, leading to a disobey of the least privilege principle. That is, the granted recipient of a privileged API is broader than intended, thus defined as an "identity confusion" in this paper. To the best of our knowledge, no prior works have studied this type of identity confusion vulnerability.

In this paper, we perform the first systematic study of identity confusion in real-world app-in-app ecosystems. We find that confusions of the aforementioned three types of identities are

widespread among all 47 studied super-apps. More importantly, such confusions lead to severe consequences such as manipulating users' financial accounts and installing malware on a smartphone. We responsibly reported all of our findings to developers of affected super-apps, and helped them to fix their vulnerabilities.

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